

Working with the Device

Davide Berti
Software Engineer
Developer Technical Services

User Accounts

The Falcon retains the notion that there is only 1 user for the device.

The creation of multiple accounts is not supported.

A root user does exist however. This account enables one to act as a system administrator via the terminal.

To log in as root enter:

```
sume
```

This allows one to mount the nfs partition, edit certain system configuration files.

If you enter a password in the system preferences this becomes the root password



Falcon Usage Guidelines

Alt 5 switches between LCD and video out.

Falcon doesn't implement the concept of On/Off.

Tap power button to turn Falcon off/places it in standby

- The application will be notified to save its state.

Tap the power button to instantly turn back on.

- Application is in the foreground

No reset key; what if an application hangs?

- "Apps" + "backspace" will kill the application.

To reboot the device hold down power button for 3 seconds.

- Falcon will reboot

Falcon Usage Guidelines

Menu emulates the control key

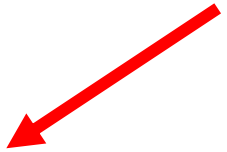
- Menu ‘c’ is equivalent to the key sequence Control ‘c’
- Menu ‘[‘ is equivalent to the escape key

Reboot and hold down the ‘b’ key to get a blob prompt

- Used to identify the HW revision of the device
- Update certain hardware specific information if required



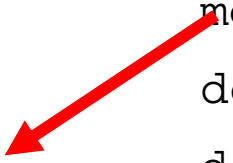
Mountpoints on the device



```
# /etc/fstab
# fstab for NAND rootfs
devfs           /dev                devfs           defaults    0 0
devpts          /dev/pts            devpts          mode=0622   0 0
proc            /proc               proc            defaults    0 0
usbfs           /proc/bus/usb       usbfs           defaults    0 0
192.168.1.100:/opt/holly/sysroot/arm-linux/ /mnt/nfs      nfs
  noauto,nolock,rsize=1024 0 0
/dev/ide/host0/bus0/target0/lun0/part1     /mnt/hda1       ext2
  rw,noatime 0 0
```

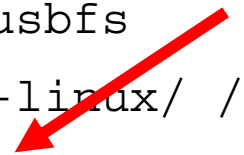
Mountpoints on the device

```
# /etc/fstab
# fstab for NAND rootfs
devfs           /dev                    devfs           defaults    0 0
devpts          /dev/pts                devpts          mode=0622   0 0
proc            /proc                   proc            defaults    0 0
usbfs           /proc/bus/usb           usbfs           defaults    0 0
192.168.1.100:/opt/holly/sysroot/arm-linux/ /mnt/nfs      nfs
  noauto,nolock,rsize=1024 0 0
/dev/ide/host0/bus0/target0/lun0/part1     /mnt/hda1       ext2
  rw,noatime 0 0
```

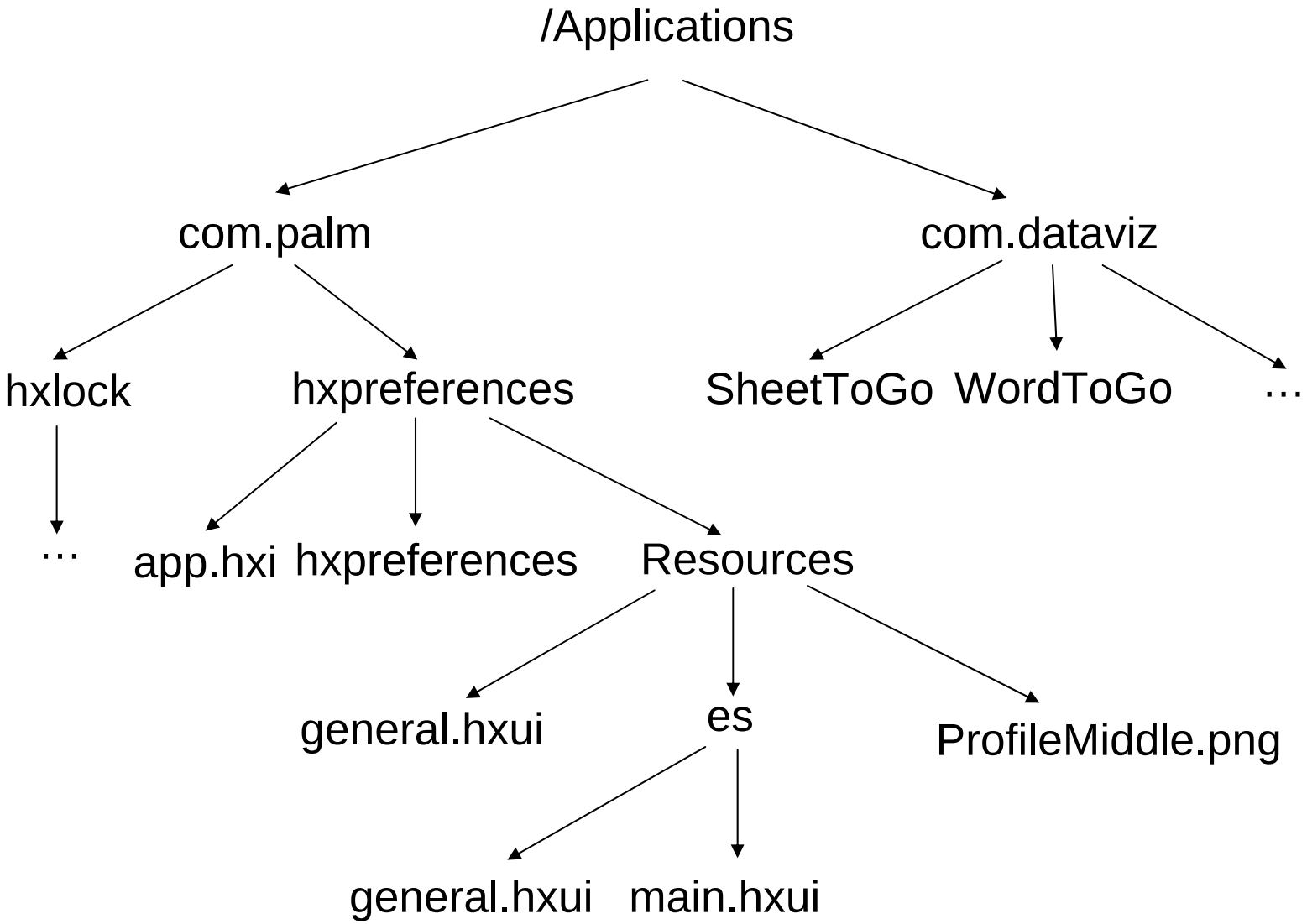


Mountpoints on the device

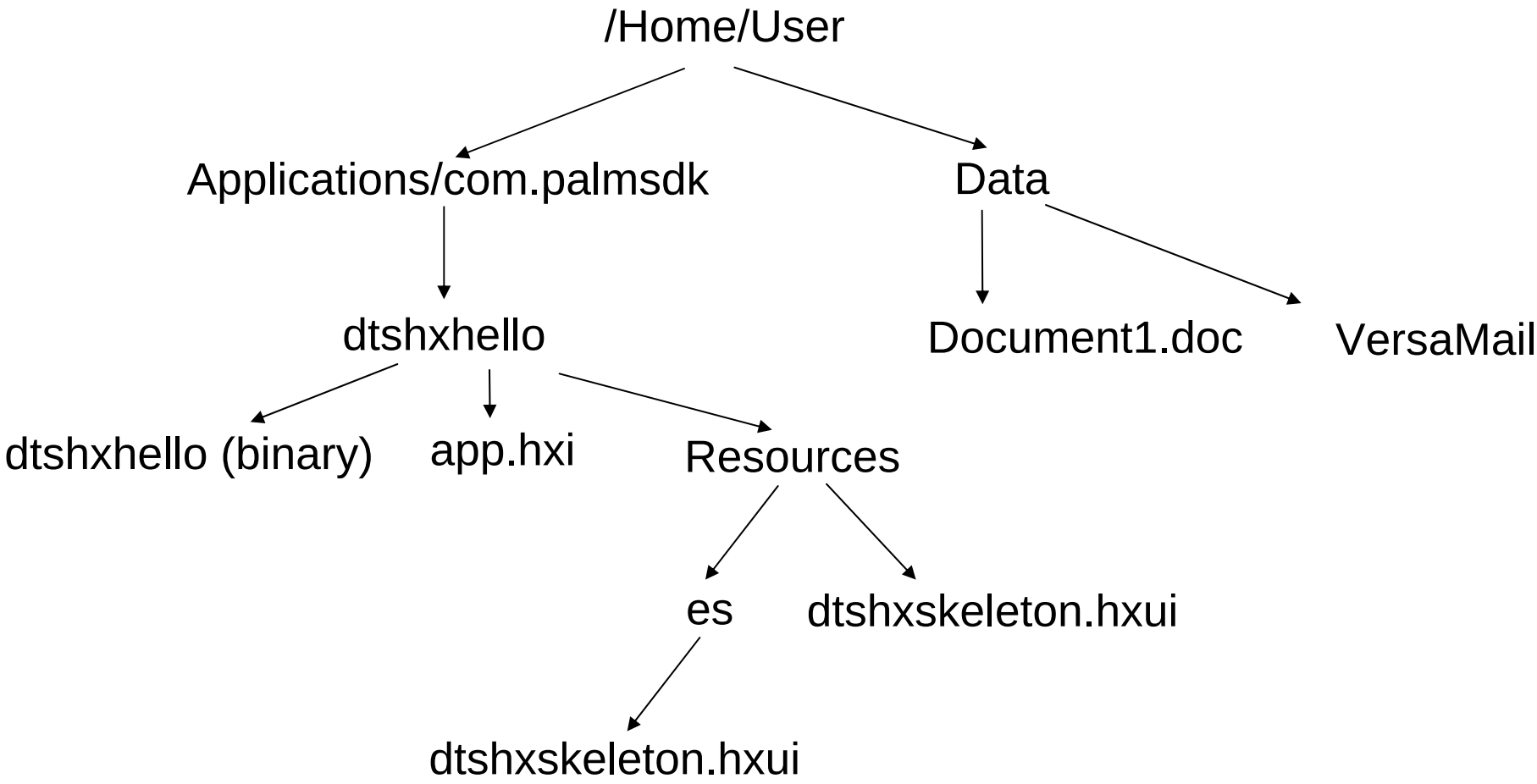
```
# /etc/fstab
# fstab for NAND rootfs
devfs           /dev                    devfs           defaults    0 0
devpts          /dev/pts                devpts          mode=0622   0 0
proc            /proc                   proc            defaults    0 0
usbfs           /proc/bus/usb           usbfs           defaults    0 0
192.168.1.100:/opt/holly/sysroot/arm-linux/ /mnt/nfs      nfs
  noauto,nolock,rsize=1024 0 0
/dev/ide/host0/bus0/target0/lun0/part1     /mnt/hda1       ext2
  rw,noatime 0 0
```



Exploring the file system: /Applications



Exploring the file system: /Home



/proc filesystem

What is /proc?

- The /proc filesystem is a direct *reflection* of the system kept in memory and represented in a hierarchal manner. The effort of the /proc filesystem is to provide an easy way to view kernel and information about currently running processes.
- Some commands (ps for example) read /proc directly to get information about the state of the system.

What /proc can do for an Administrator?

- Viewing Statistical Information
- Finding out Hardware Information
- Modifying Runtime Parameters
- Viewing and Modifying Network and Host Parameters
- Memory and Performance Information
- There are some things to take note of, most of those tasks can be done with tools that either peruse /proc or query the kernel directly.



/proc filesystem ..

...
715
721
722
723
audio
battery
bus
cmdline
cpu
cpuinfo
devices
driver
...



/etc filesystem

Hosts system-wide specific configuration files

- btprot
- busybox.conf
- cron.d
- deadlockd.conf
- directfbrc
- dropbear
- eth.conf
- eth.conf.dhcp
- fb_marathon.modes
- fb.modes
- fstab
- group
- hostname
- hosts
- hotplug



Network configuration - ifconfig

Ifconfig is used to configure the kernel-resident network interfaces. It is used at boot time to set up interfaces as necessary. After that, it is usually only needed when debugging or when system tuning is needed.

If no arguments are given, ifconfig displays the status of the currently active interfaces. If a single interface argument is given, it displays the status of the given interface only; if a single -a argument is given, it displays the status of all interfaces, even those that are down. Otherwise, it configures an interface.



Network configuration - ifconfig

```
eth0   Link encap:Ethernet  HWaddr 00:40:F4:D1:AF:63
       inet addr:192.168.1.102  Bcast:192.168.1.255  Mask:255.255.255.0
       UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
       RX packets:56 errors:0 dropped:0 overruns:0 frame:0
       TX packets:53 errors:0 dropped:0 overruns:0 carrier:0

lo     Link encap:Local Loopback
       inet addr:127.0.0.1  Mask:255.0.0.0
       UP LOOPBACK RUNNING  MTU:16436  Metric:1
       RX packets:0 errors:0 dropped:0 overruns:0 frame:0
```



Linux Essentials: Logging – kernel

Log messages from the kernel.

dmesg is used to examine or control the kernel ring buffer. To inspect the last 100 lines in the ring buffer. From the terminal on the device or workstation enter:

```
dmesg | tail -n 100 | more
```



Linux Essentials: Logging – syslog-ng

syslog-ng

Syslog entries contain a timestamp and a text string. They are rotated out and backed up on a regular basis to syslog.x.gz

To inspect this file from a terminal enter:

```
more /var/log/syslog
```

To create an entry in your program enter

```
syslog(LOG_INFO, "%s: Application terminated
  cleanly.\n", __FUNCTION__);
```



Linux Essentials: Logging – hxtek

Hxtek runs as a daemon. It pipes stdin/stdout/stderr to /dev/null.

What does this mean?

An application running in the simulator will not be able to see printf messages on the console. How to enable these log messages…



Linux Essentials: Logging – hxtek

On the device, console output for hxtek can be enabled in this manner. From a terminal enter:

```
enable_stderr_logging
```

Reboot the device. Messages appear in /var/log/hxtek

In the simulator, console output for hxtek can be enabled in this manner. Launch hxtek with the –o flag..

```
/opt/holly/sysroot/i386-linux/usr/bin/hxtek –o
```

Or via bash functions..

```
dohxtek -o
```



Linux Essentials: Logging – HX

hxdebug.h includes a set of macros that can be used to post information into var/log/hxtek.

In your application please define –DHX_DEBUG in the Makefile or in a C file.

```
#define HX_INFO(...)

#define HX_WARNING(...)

#define HX_ERROR_MESSAGE(...)

#define HX_ASSERT(exp)   do {} while (0)

#define HX_MESSAGE_IF_ERROR(exp, pErrorString)  do {} while
   (0)

#define HX_INFO_AT(...)

#define HX_WARNING_AT(...)

#define HX_ERROR_MESSAGE_AT(...)

#define HX_EXIT_IF_ERROR(exp, errorVariable, label)
```



Linux Essentials: Logging – DirectFB



- Examining DirectFB log messages
  - Flash the device with a debug ROM
  - Edit the file /etc/directfbrc and comment/uncomment appropriately

Core Files

Use Compact Flash to allow for larger core dump files

Core dumps can be quite large. It is recommended, but optional, to integrate a CF card into the system.

Enable core-dumps from the terminal

- To enable coredumps, run:
  - sume
  - /usr/bin/enable_coredumps # enable core dumps
- To disable coredumps, run:
  - sume
  - /usr/bin/enable_coredumps disable # disable core dumps



Core Files

Cores are written to /cores/processname.time.pid.core

An example:

Launch the filemanager application.

Force Quit (Apps + Backspace)

Examine /cores

/cores/filemanager.28923.290.core.

Copy over nfs to the development workstation.



Minicores - explained

minicore is a mechanism for the kernel to dump stacktrace information in a compact format when an application crashes. minicore can also generate a stacktrace for select events such as the detection of potential deadlocks.

These minicores give stack trace information.

In order to get a pretty stacktrace of an application crash, you must include symbols and frame pointer with your executable.

.



Minicores - prerequisites

Stripping the symbols from the binary.

From a terminal:

```
/opt/holly/toolroot/bin/arm-linux-strip -g binary
```

- frame pointer (**-fno-omit-frame-pointer**)
- In order to get a pretty stacktrace of an application crash, you must have include symbols and frame pointer with your executable.



Minicores

A GUI is spawned when a minicore file is detected. The user at this point has several choices.

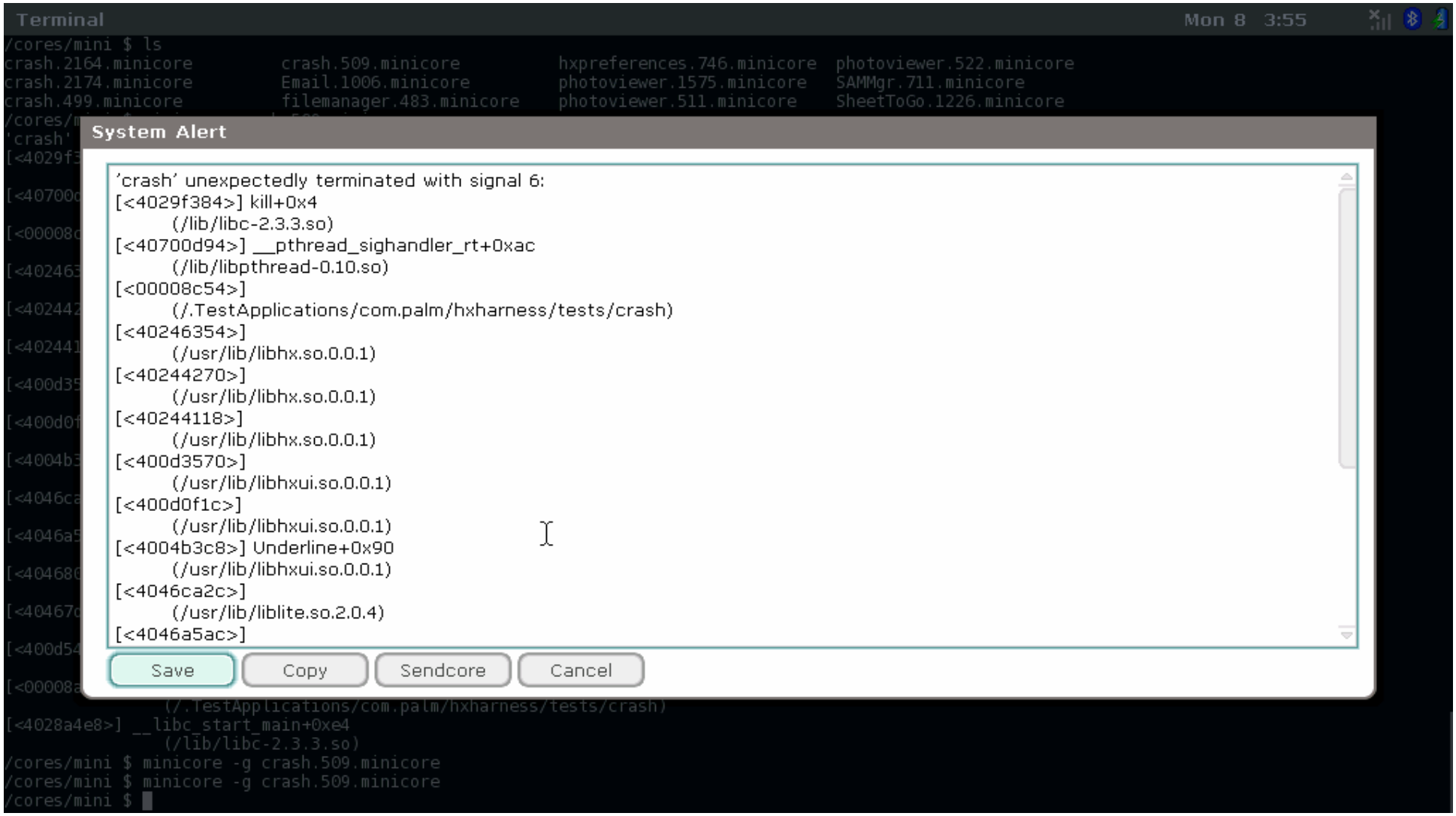
In case you missed the crash dialog, a file is also generated called /cores/mini/app_name.pid.minicore. To obtain a stacktrace on the device, you only need to run:

minicore /cores/mini/app_name.pid.minicore. To access the gui again, you may do

minicore -g /cores/mini/app_name.pid.minicore



Minicores


Terminal
Mon 8 3:55
/cores/mini $ ls
crash.2164.minicore
crash.2174.minicore
crash.499.minicore
crash.509.minicore
Email.1006.minicore
filemanager.483.minicore
hxpreferences.746.minicore
photoviewer.1575.minicore
photoviewer.511.minicore
photoviewer.522.minicore
SAMMgr.711.minicore
SheetToGo.1226.minicore
System Alert
'crash' unexpectedly terminated with signal 6:
[<4029f384>] kill+0x4
(/lib/libc-2.3.3.so)
[<40700d94>] __pthread_sighandler_rt+0xac
(/lib/libpthread-0.10.so)
[<00008c54>]
(/.TestApplications/com.palm/hxharness/tests/crash)
[<40246354>]
(/usr/lib/libhx.so.0.0.1)
[<40244270>]
(/usr/lib/libhx.so.0.0.1)
[<40244118>]
(/usr/lib/libhx.so.0.0.1)
[<400d3570>]
(/usr/lib/libhxui.so.0.0.1)
[<400d0f1c>]
(/usr/lib/libhxui.so.0.0.1)
[<4004b3c8>] Underline+0x90
(/usr/lib/libhxui.so.0.0.1)
[<4046ca2c>]
(/usr/lib/liblite.so.2.0.4)
[<4046a5ac>]
Save
Copy
Sendcore
Cancel
(/.TestApplications/com.palm/hxharness/tests/crash)
[<4028a4e8>] __libc_start_main+0xe4
(/lib/libc-2.3.3.so)
/cores/mini $ minicore -g crash.509.minicore
/cores/mini $ minicore -g crash.509.minicore
/cores/mini $

The Sendcore Option

Currently, sendcore is configured to send corefiles to a Palm ftp server.



Linux References…

- **Linux in a Nutshell, Fourth Edition (Paperback)**
  by Ellen Siever, Aaron Weber, Stephen Figgins (Editor)
- **Running Linux, Fourth Edition (Paperback)**
  by Matt Welsh, Lar Kaufman, Matthias Kalle Dalheimer, Terry Dawson
- **Building Embedded Linux Systems (Paperback)**
  by Karim Yaghmour
- **Learning the bash Shell (Nutshell Handbooks) (Paperback)**
  by Cameron Newham, Bill Rosenblatt
- **Linux Device Drivers, 3rd Edition (Paperback)**
  by Jonathan Corbet, Alessandro Rubini, Greg Kroah-Hartman

